Problem M2: Rapidity in an Asymmetric Collider

Worked Solution — Corrected Version (General Masses in Part a)

Midterm Preparation Notes

Preamble: Key Definitions

Rapidity y

For a particle with energy E and longitudinal momentum p_z:

$$y = \frac{1}{2}\ln\frac{E+p_z}{E-p_z}, \qquad \tanh y = \frac{p_z}{E}, \qquad \beta = \tanh y \tag{1}$$

Rapidity Shift Under Boosts

Under a Lorentz boost along z by rapidity ϕ, every particle's rapidity shifts by a constant:

$$y' = y - \phi \tag{2}$$

Proof: Under a boost, $E' + p'_z = e^{-\phi}(E+p_z)$ and $E' - p'_z = e^{+\phi}(E-p_z)$, so:

$$y' = \frac{1}{2}\ln\frac{E'+p'_z}{E'-p'_z} = \frac{1}{2}\ln\frac{e^{-\phi}(E+p_z)}{e^{+\phi}(E-p_z)} = y - \phi$$

CM Frame

The center-of-mass frame is defined as the frame where the total three-momentum vanishes: $\vec{p}'_{\text{tot}} = 0$. Equivalently, it is the **rest frame of the total four-momentum** $P^\mu = \sum p_i^\mu$. In this frame, $P^\mu = (\sqrt{s}, \vec{0})$.

1 Part (a): $y_{\mathbf{CM}} = y_{\mathbf{tot}}$ — General Proof ($m_a \neq m_b$)

What the question asks

Two particles a and b with **possibly different masses** m_a and m_b collide along the z-axis. We must show that the rapidity of the CM frame (y_{CM}, i.e. the boost rapidity needed to zero out total p_z) equals the rapidity computed from the total four-momentum $P^\mu = p_a^\mu + p_b^\mu$.

These are *a priori* different objects:

- $y_{\mathbf{CM}}$: a property of a *frame transformation* (what boost zeros the total momentum?)
- $y_{\mathbf{tot}}$: a property of a *four-vector* (apply the rapidity formula to P^μ)

We show they are equal.

Step 1: Find $y_{\mathbf{CM}}$ by demanding $p'_z = 0$

The total four-momentum in the lab is:

$$P^\mu = p_a^\mu + p_b^\mu = (E_a + E_b,\ 0,\ 0,\ p_{z,a} + p_{z,b}) \equiv (E_{\mathrm{tot}},\ 0,\ 0,\ p_{z,\mathrm{tot}}) \tag{3}$$

We do **not** assume anything about the masses — E_a, E_b, $p_{z,a}$, $p_{z,b}$ are completely general. Now boost along z by rapidity ϕ. The boosted z-component of total momentum is:

$$p'_{z,\mathrm{tot}} = \gamma_\phi (p_{z,\mathrm{tot}} - \beta_\phi\, E_{\mathrm{tot}}) \tag{4}$$

Set this to zero (the CM condition):

$$p'_{z,\mathrm{tot}} = 0 \quad \Longrightarrow \quad p_{z,\mathrm{tot}} = \beta_\phi\, E_{\mathrm{tot}} \quad \Longrightarrow \quad \beta_\phi = \frac{p_{z,\mathrm{tot}}}{E_{\mathrm{tot}}} \tag{5}$$

Since $\tanh\phi = \beta_\phi$:

$$\tanh(y_{\mathrm{CM}}) = \frac{p_{z,\mathrm{tot}}}{E_{\mathrm{tot}}} \tag{6}$$

Converting to the logarithmic form using $\tanh^{-1}(x) = \frac{1}{2}\ln\frac{1+x}{1-x}$:

$$y_{\mathrm{CM}} = \frac{1}{2}\ln\frac{1 + p_{z,\mathrm{tot}}/E_{\mathrm{tot}}}{1 - p_{z,\mathrm{tot}}/E_{\mathrm{tot}}} = \frac{1}{2}\ln\frac{E_{\mathrm{tot}} + p_{z,\mathrm{tot}}}{E_{\mathrm{tot}} - p_{z,\mathrm{tot}}} \tag{7}$$

Step 2: Compute the rapidity of P^μ

By the *definition* of rapidity applied to any four-vector with energy E and z-momentum p_z:

$$y_{\mathrm{tot}} = \frac{1}{2}\ln\frac{E_{\mathrm{tot}} + p_{z,\mathrm{tot}}}{E_{\mathrm{tot}} - p_{z,\mathrm{tot}}} \tag{8}$$

Step 3: Compare

Equations (7) and (8) are identical.

Part (a) Result — Fully General

For **any** two particles a and b (any masses, any energies), the rapidity of the CM frame equals the rapidity of the total four-momentum:

$$\boxed{y_{\mathrm{CM}} = y_{\mathrm{tot}} = \frac{1}{2}\ln\frac{E_{\mathrm{tot}} + p_{z,\mathrm{tot}}}{E_{\mathrm{tot}} - p_{z,\mathrm{tot}}}} \tag{9}$$

No assumptions about masses were used.

Why this is almost tautological

The rapidity of a four-vector *is* the boost rapidity that takes it to rest. The CM frame *is* the rest frame of P^μ. So $y_{\text{CM}} = y_{\text{tot}}$ is really just saying "the boost needed to bring P^μ to rest is the rapidity of P^μ" — which is the definition of rapidity.
The value of this result is not that it's deep, but that it gives you a **practical recipe**: to find the CM boost, just compute E_{tot} and $p_{z,\text{tot}}$ and plug into the rapidity formula. No need to solve kinematic equations.

Alternative route the problem suggests

The problem says "one way is to explicitly transform p_z for a and b to the CM, and impose the required relation between p_a and p_b there." This means:

1. Boost both $p_{z,a}$ and $p_{z,b}$ by unknown rapidity ϕ
2. In the CM frame, $|\vec{p}'_a| = |\vec{p}'_b|$ and they're back-to-back, so $p'_{z,a} + p'_{z,b} = 0$
3. Solve for ϕ

That's exactly what we did in Step 1, but phrased in terms of the *individual* momenta rather than the total. The result is the same because $p'_{z,a} + p'_{z,b} = p'_{z,\text{tot}} = \gamma(p_{z,\text{tot}} - \beta E_{\text{tot}}) = 0$ gives the same $\beta = p_{z,\text{tot}}/E_{\text{tot}}$ regardless.

2 Part (b): Equal Masses $\Rightarrow y_{\mathrm{CM}} = \frac{1}{2}(y_a + y_b)$

What the question asks

Specialize to $m_a = m_b \equiv m$ and show that $y_{\mathrm{CM}} = \frac{1}{2}(y_a + y_b)$.

Now we use the hyperbolic parameterization

With equal masses and $p_T = 0$, the beam four-momenta are:

$$p_a^\mu = m(\cosh y_a,\ 0,\ 0,\ \sinh y_a) \tag{10}$$
$$p_b^\mu = m(\cosh y_b,\ 0,\ 0,\ \sinh y_b) \tag{11}$$

(The sign of the z-component is encoded in $y_b < 0$ for a beam moving in $-z$.)

Total:

$$E_{\mathrm{tot}} = m(\cosh y_a + \cosh y_b) \tag{12}$$
$$p_{z,\mathrm{tot}} = m(\sinh y_a + \sinh y_b) \tag{13}$$

Apply the result from part (a):

$$y_{\mathrm{CM}} = \frac{1}{2}\ln\frac{E_{\mathrm{tot}} + p_{z,\mathrm{tot}}}{E_{\mathrm{tot}} - p_{z,\mathrm{tot}}} = \frac{1}{2}\ln\frac{m(\cosh y_a + \cosh y_b) + m(\sinh y_a + \sinh y_b)}{m(\cosh y_a + \cosh y_b) - m(\sinh y_a + \sinh y_b)} \tag{14}$$

The mass m cancels (**this is where equal masses matters** — both terms have the same m):

$$y_{\mathrm{CM}} = \frac{1}{2}\ln\frac{(\cosh y_a + \sinh y_a) + (\cosh y_b + \sinh y_b)}{(\cosh y_a - \sinh y_a) + (\cosh y_b - \sinh y_b)} = \frac{1}{2}\ln\frac{e^{y_a} + e^{y_b}}{e^{-y_a} + e^{-y_b}} \tag{15}$$

Now factor. **Numerator:** $e^{y_a}+e^{y_b} = e^{(y_a+y_b)/2}\left(e^{(y_a-y_b)/2}+e^{-(y_a-y_b)/2}\right) = 2e^{(y_a+y_b)/2}\cosh\frac{y_a-y_b}{2}$.
Denominator: $e^{-y_a} + e^{-y_b} = 2e^{-(y_a+y_b)/2}\cosh\frac{y_a-y_b}{2}$.
The cosh factors cancel:

$$y_{\mathrm{CM}} = \frac{1}{2}\ln\frac{e^{(y_a+y_b)/2}}{e^{-(y_a+y_b)/2}} = \frac{1}{2}\ln e^{y_a+y_b} = \frac{1}{2}(y_a + y_b) \tag{16}$$

Part (b) Result

$$\boxed{y_{\mathrm{CM}} = \frac{1}{2}(y_a + y_b) \qquad (m_a = m_b)} \tag{17}$$

For a symmetric collider ($E_a = E_b$): $y_a = -y_b$, so $y_{\mathrm{CM}} = 0$ (lab = CM).
For an asymmetric collider: $y_{\mathrm{CM}} \neq 0$ and the CM is boosted along z.

Why equal masses is essential

If $m_a \neq m_b$, the total four-momentum would be:

$$E_{\mathrm{tot}} + p_{z,\mathrm{tot}} = m_a e^{y_a} + m_b e^{y_b}$$

and the masses **don't** cancel in the ratio. You'd get:

$$y_{\mathrm{CM}} = \frac{1}{2}\ln\frac{m_a e^{y_a} + m_b e^{y_b}}{m_a e^{-y_a} + m_b e^{-y_b}}$$

which is **not** $\frac{1}{2}(y_a + y_b)$ in general. The clean "average" formula is specific to equal-mass beams.

3 Part (c): Beam Rapidities in the CM Frame

What the question asks

Show that in the CM frame, the two beams have rapidities $\pm\frac{1}{2}(y_a - y_b)$.

Derivation

Use the additive property (2) with $\phi = y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$:

Beam a:

$$y_a^* = y_a - y_{\text{CM}} = y_a - \frac{1}{2}(y_a + y_b) = +\frac{1}{2}(y_a - y_b) \tag{18}$$

Beam b:

$$y_b^* = y_b - y_{\text{CM}} = y_b - \frac{1}{2}(y_a + y_b) = -\frac{1}{2}(y_a - y_b) \tag{19}$$

Part (c) Result

$$\boxed{y_a^* = +\frac{1}{2}(y_a - y_b), \qquad y_b^* = -\frac{1}{2}(y_a - y_b)} \tag{20}$$

The beams are **equal and opposite** in the CM frame, as required by the definition of the CM frame for equal-mass particles.

Sanity checks

- **Symmetric collider** ($y_a = -y_b = Y$): $y_a^* = Y$, $y_b^* = -Y$. Same as lab — correct, since lab = CM.
- **Fixed target** ($y_b = 0$): $y_a^* = y_a/2$, $y_b^* = -y_a/2$. Each beam gets half the rapidity — the CM splits the difference.
- **Sign check**: Since beam a moves in $+z$ and beam b in $-z$, we have $y_a > 0 > y_b$, so $y_a - y_b > 0$, giving $y_a^* > 0$ and $y_b^* < 0$. ✓

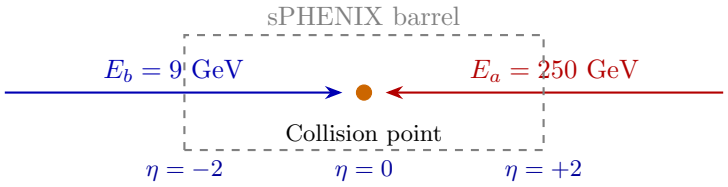
4 Part (d): RHIC Asymmetric Run and sPHENIX Acceptance

The setup

RHIC runs asymmetric proton-proton collisions:

- Beam a: $E_a = 250$ GeV (moving in $+z$, the maximum RHIC energy)
- Beam b: $E_b = 9$ GeV (moving in $-z$, the minimum "booster" injection energy)
- Proton mass: $m_p = 0.9383$ GeV

The purpose: shift physics away from mid-rapidity to study forward/gluon-dominated regimes.



Step 1: Compute $\sqrt{s}$

For beam particles with $E \gg m$, $|\vec{p}| \approx E$. The four-momenta are:

$$p_a^\mu \approx (E_a, 0, 0, +E_a), \qquad p_b^\mu \approx (E_b, 0, 0, -E_b) \tag{21}$$

$$s = (p_a + p_b)^2 = (E_a + E_b)^2 - (E_a - E_b)^2 = 4E_aE_b \tag{22}$$

$\sqrt{s}$

$$\boxed{\sqrt{s} = 2\sqrt{E_aE_b} = 2\sqrt{250 \times 9} = 2\sqrt{2250} \approx 94.9 \text{ GeV}} \tag{23}$$

Step 2: Compute y_a and y_b

For a beam particle moving in $+z$ with $E \gg m$:

$$y_a \approx \ln \frac{2E_a}{m_p} = \ln \frac{500}{0.9383} = \ln(532.9) \approx 6.28 \tag{24}$$

For beam b moving in $-z$ ($p_z = -E_b$):

$$y_b \approx -\ln \frac{2E_b}{m_p} = -\ln \frac{18}{0.9383} = -\ln(19.18) \approx -2.95 \tag{25}$$

Beam rapidities

$$\boxed{y_a \approx +6.28, \qquad y_b \approx -2.95} \tag{26}$$

Step 3: Compute y_{CM}

Using the part (b) result (both beams are protons, so $m_a = m_b$):

$$y_{\text{CM}} = \frac{1}{2}(y_a + y_b) = \frac{1}{2}(6.28 - 2.95) = \frac{1}{2}(3.33) \tag{27}$$

y_{CM}

$$\boxed{y_{\text{CM}} \approx +1.66} \tag{28}$$

The CM frame is boosted by $y = 1.66$ in the $+z$ direction (chasing the 250 GeV beam).

Cross-check: $y_{\text{CM}} = \frac{1}{2}\ln\frac{E_a}{E_b} = \frac{1}{2}\ln\frac{250}{9} = \frac{1}{2}(3.325) = 1.66$ ✓
(This shortcut works because for ultra-relativistic beams, $y_a + y_b \approx \ln(2E_a/m) - \ln(2E_b/m) = \ln(E_a/E_b)$.)

Step 4: Translate the sPHENIX acceptance

Symmetric case (reference)

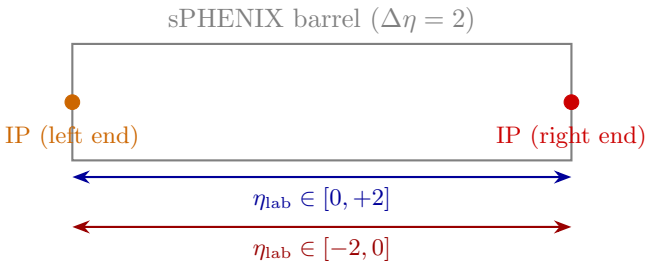
In normal symmetric running (250 + 250), collisions occur at the barrel center. sPHENIX Central Arm covers:

$$-1 < \eta < +1 \qquad (\text{lab} = \text{CM})$$

Asymmetric case: collision at each end of the barrel

The problem states:

- The barrel has a fixed geometric extent of $\Delta\eta = 2$.
- The collision point is moved to either *end* of the barrel.
- The acceptance is $[0, +2]$ at one end and $[-2, 0]$ at the other (in the lab).



Key assumption: $\eta = y$

The problem states this explicitly. For the detected particles (pions, etc.), $m \ll p$, so pseudorapidity ≈ rapidity. Therefore, to convert lab-frame η to CM-frame y^*:

$$y^* = \eta_{\text{lab}} - y_{\text{CM}} \tag{29}$$

Why $\eta = y$ justifies this shift

This is a subtle point worth pausing on. The additive property (2) says *rapidity* shifts by a constant under boosts. Pseudorapidity does **not** shift by a constant in general — it transforms in a more complicated way because it depends on θ, not on E and p_z.

But if $\eta \approx y$ for the particles we're detecting, then the shift formula works for η too:

$$y^*_{\text{CM}} = y_{\text{lab}} - y_{\text{CM}} \approx \eta_{\text{lab}} - y_{\text{CM}}$$

This is what the problem means by "make sure you justify why $\eta = y$."

The translation

Collision at left end ($\eta_{\text{lab}} \in [0, +2]$):

$$y^* \in [0 - 1.66,\ +2 - 1.66] = [-1.66,\ +0.34] \tag{30}$$

Collision at right end ($\eta_{\text{lab}} \in [-2, 0]$):

$$y^* \in [-2 - 1.66,\ 0 - 1.66] = [-3.66,\ -1.66] \tag{31}$$

Part (d) — All Results

$\sqrt{s}$	94.9 GeV
y_a	+6.28
y_b	−2.95
y_{CM}	+1.66
Left-end acceptance (CM frame)	$\boxed{y^* \in [-1.66,\ +0.34]}$
Right-end acceptance (CM frame)	$\boxed{y^* \in [-3.66,\ -1.66]}$

Why bother? The physics motivation

In symmetric $250 + 250$ running, sPHENIX sees $y^* \in [-1, +1]$ — mid-rapidity, where both beams contribute roughly equally and the physics is well understood.
By going asymmetric $(250 + 9)$ and sliding the collision point:

- One configuration reaches $y^* \approx -3.7$: deep into the **forward region** of the low-energy beam. In this regime, the partons probed in the 9 GeV proton have large momentum fractions ($x \to 1$), while partons in the 250 GeV proton have very small x. This is the **gluon saturation** regime — the whole reason RHIC does this.
- The other configuration ($y^* \in [-1.66, +0.34]$) probes intermediate rapidities, complementing the forward measurement.

All of this is achieved **without building a new detector** — just by choosing beam energies and collision point locations. That's the power of understanding rapidity.

Complete Summary

Part	Result	Key Step
(a)	$y_{\text{CM}} = y_{\text{tot}} = \frac{1}{2}\ln\frac{E_{\text{tot}} + p_{z,\text{tot}}}{E_{\text{tot}} - p_{z,\text{tot}}}$ (general, any masses)	Boost P^μ, set $p'_z = 0$, compare with rapidity defini
(b)	$y_{\text{CM}} = \frac{1}{2}(y_a + y_b)$ (equal masses only)	m cancels, use $\cosh y + \sinh y = e^y$
(c)	$y^*_{a,b} = \pm\frac{1}{2}(y_a - y_b)$	Subtract y_{CM} from each beam rapidity
(d)	$\sqrt{s} \approx 94.9$ GeV, $y_{\text{CM}} \approx 1.66$ Acceptance: $[-1.66, +0.34]$ and $[-3.66, -1.66]$	$s = 4E_aE_b$, shift η by y_{CM}

The Three Big Ideas

1. **Part (a)**: The CM boost rapidity equals the rapidity of the total four-momentum — true for any masses, any energies. This is essentially the *definition* of rapidity: it's the boost parameter of a four-vector's rest frame.
2. **Parts (b)–(c)**: For equal-mass beams, y_{CM} is just the average, and the CM-frame rapidities are symmetric. This follows from the additive property of rapidity, which is the reason rapidity exists as a concept in collider physics.
3. **Part (d)**: By choosing asymmetric beam energies, you *slide* the CM-frame acceptance window of a fixed detector to probe new rapidity regions — a practical trick used at RHIC, and analogous to the asymmetric B-factory strategy used at PEP-II (9 GeV e^- on 3.1 GeV e^+) to resolve B-meson decay vertices.